

KOSLAN, Russian Federation

WMO: 226950

Lat: 63.474N

Lon: 48.845E

Elev: 122

StdP: 99.86

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.0	-35.8	0.1	-32.8	-32.8	0.2	-29.9	8.2	-10.2	7.4	-11.4	1.6	80	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	27.9	18.6	25.5	17.4	23.4	16.6	19.8	25.4	18.7	23.6	17.5	21.9	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.0	21.7	16.8	12.1	21.1	15.7	11.3	19.9	57.2	25.3	53.6	23.8	49.9	21.9	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.8	6.2	DB	-35.6	30.0	4.8	2.1	-39.0	31.5	-41.8	32.8	-44.5	34.0	-47.9	35.5	(4)
(5)			WB	-35.7	21.3	4.7	1.7	-39.1	22.5	-41.9	23.5	-44.5	24.4	-48.0	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-15.0	-11.6	-6.5	0.5	7.0	13.0	16.4	13.2	7.6	1.2	-5.9	-11.1	(6)	
(7)		DBStd	11.90	8.11	7.83	5.63	4.71	5.80	4.82	4.24	4.17	3.69	3.97	6.55	8.68	(7)	
(8)		HDD10.0	3843	775	606	511	287	133	22	3	14	88	273	477	656	(8)	
(9)		HDD18.3	6449	1033	839	769	536	353	171	88	168	321	531	727	914	(9)	
(10)		CDD10.0	487	0	0	0	1	40	113	202	114	17	1	0	0	(10)	
(11)		CDD18.3	51	0	0	0	0	2	12	28	8	0	0	0	0	(11)	
(12)		CDH23.3	487	0	0	0	0	31	124	269	62	1	0	0	0	(12)	
(13)		CDH26.7	118	0	0	0	0	5	29	68	16	0	0	0	0	(13)	
(14)	Wind	WSAvg	3.1	2.9	3.1	3.2	3.4	3.4	3.2	2.8	2.7	2.8	3.1	3.1	2.9	(14)	
(15)	Precipitation	PrecAvg	604	39	33	30	37	44	65	60	80	58	66	52	42	(15)	
(16)		PrecMax	701	53	74	52	62	87	130	92	141	114	120	104	65	(16)	
(17)		PrecMin	485	19	12	2	16	13	19	12	37	23	23	10	19	(17)	
(18)		PrecStd	68	10	15	14	14	17	32	24	28	21	27	24	13	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	1.6	6.9	17.1	27.1	29.8	31.1	28.4	20.8	12.5	5.9	1.1	(19)	
(20)		MCWB	0.0	0.6	2.4	9.0	15.5	19.3	20.1	19.5	15.0	10.6	5.1	0.6	0.6	(20)	
(21)		2%	DB	-0.9	0.3	3.6	12.0	22.8	26.7	28.6	24.6	17.4	9.6	3.2	0.4	(21)	
(22)		MCWB	-1.2	-0.1	0.7	6.1	13.6	17.6	19.2	18.0	13.2	8.6	3.0	0.1	0.1	(22)	
(23)		5%	DB	-2.3	-0.7	1.9	9.0	19.9	24.2	26.2	22.1	15.2	7.8	1.8	-0.1	(23)	
(24)		MCWB	-2.7	-1.2	-0.1	4.0	11.9	16.1	18.1	16.8	12.1	6.9	1.6	-0.3	-0.3	(24)	
(25)		10%	DB	-4.4	-2.0	0.8	7.0	17.0	21.6	24.2	19.9	13.3	6.3	0.8	-1.0	(25)	
(26)		MCWB	-4.8	-2.4	-0.6	3.2	10.6	15.2	17.2	15.7	11.0	5.7	0.5	-1.3	-1.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.9	2.9	9.7	16.8	20.4	21.8	20.9	15.5	11.2	5.6	0.7	(27)	
(28)		MDCB	0.3	1.3	5.9	16.1	24.5	26.8	28.1	27.4	19.8	12.1	5.8	0.8	0.8	(28)	
(29)		2%	WB	-1.2	-0.1	1.4	6.8	15.0	19.0	20.2	19.0	13.8	8.8	3.0	0.2	(29)	
(30)		MDCB	-0.9	0.2	2.8	11.1	21.9	24.6	26.2	23.2	16.5	9.5	3.2	0.3	0.3	(30)	
(31)		5%	WB	-2.6	-1.2	0.4	4.8	12.9	17.5	19.2	17.7	12.5	7.0	1.7	-0.3	(31)	
(32)		MDCB	-2.3	-0.8	1.6	7.9	19.0	22.5	24.4	20.9	14.7	7.6	1.8	-0.2	-0.2	(32)	
(33)		10%	WB	-4.8	-2.4	-0.4	3.5	11.0	16.0	18.1	16.2	11.2	5.8	0.6	-1.3	(33)	
(34)		MDCB	-4.6	-2.1	0.7	6.1	16.1	20.8	22.8	19.2	13.0	6.4	0.7	-1.0	-1.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.2	6.1	8.0	8.0	9.4	9.4	9.7	8.1	6.4	3.8	4.1	5.4	5.4	(35)	
(36)		5% DB	MCDBR	5.2	3.9	7.8	11.8	13.5	13.2	12.9	11.2	10.6	5.8	4.0	3.4	3.4	(36)
(37)		MCWBR	5.2	3.6	5.7	7.3	7.2	6.1	5.5	5.5	6.6	4.7	3.9	3.5	3.5	(37)	
(38)		5% WB	MCDBR	5.4	3.7	5.7	9.8	12.6	11.5	11.3	10.3	9.2	5.2	3.9	3.4	3.4	(38)
(39)		MCWBR	5.4	3.5	4.6	6.8	7.3	6.4	5.7	5.8	6.2	4.5	3.9	3.5	3.5	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.232	0.282	0.314	0.348	0.347	0.352	0.398	0.366	0.331	0.304	0.234	0.218	0.218	(40)	
(41)		taud	1.948	2.060	2.090	2.132	2.415	2.474	2.345	2.415	2.502	2.351	2.022	1.925	1.925	(41)	
(42)		Ebn at Noon	346	621	761	820	867	869	809	806	770	620	343	155	155	(42)	
(43)		Edh at Noon	32	68	100	120	100	96	107	91	69	52	28	15	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.19	0.85	2.21	3.97	4.93	5.25	5.24	3.57	1.90	0.73	0.20	0.08	0.08	(44)	
(45)		RadStd	0.02	0.16	0.32	0.39	0.35	0.53	0.44	0.31	0.21	0.10	0.04	0.01	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page